

Descriptive Statistics Project

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Project:

Descriptive statistics of the daily average temperature from 02/01/2025 to 02/10/2025 in Ashland, VA.

Direct Webpage: <https://www.weather.gov/wrh/Climate?wfo=akq>

Dataset: Values are displayed in °F.

NOWData - NC	
Climate	
Click column	
Date	Maximum
2025-02-01	56
2025-02-02	44
2025-02-03	67
2025-02-04	71
2025-02-05	49
2025-02-06	48
2025-02-07	62
2025-02-08	50
2025-02-09	53
2025-02-10	47

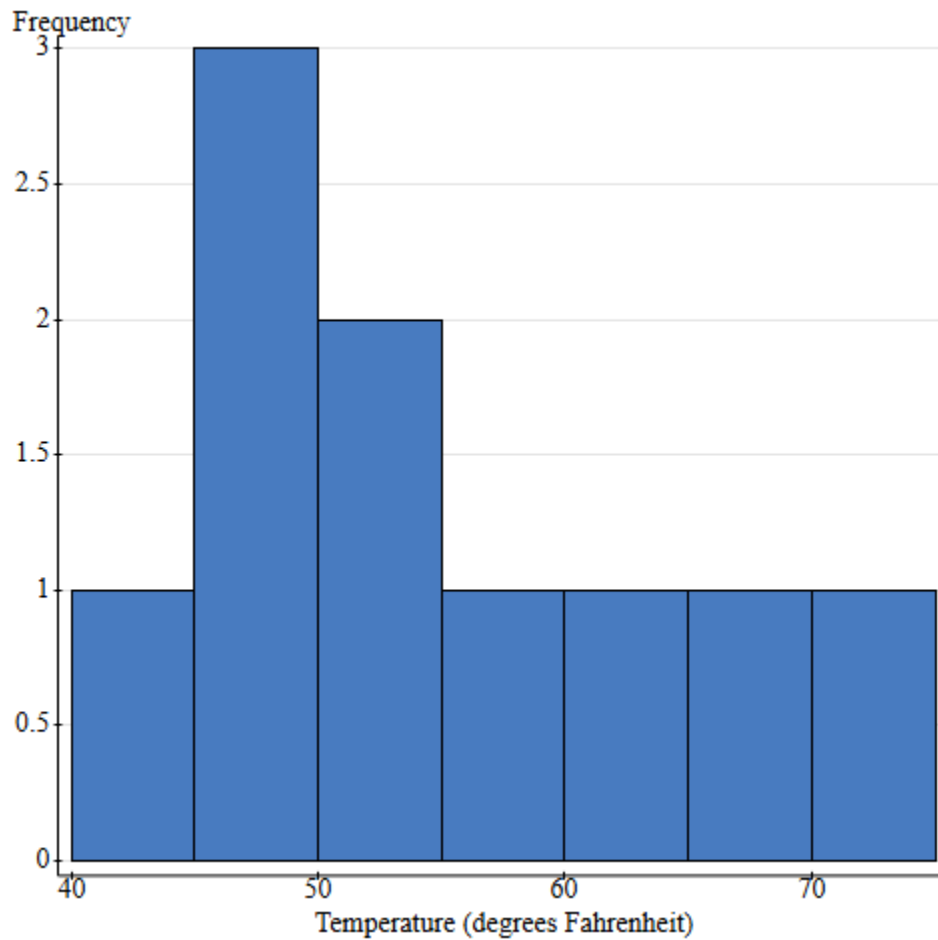
Values: 56,44,67,71,49,48,62,50,53,47

Objectives:

- 1) Data Presentation: The dataset will be presented using a histogram and I will briefly describe it.
- 2) Measures of center: I will calculate and interpret the mean, median, mode, and midrange of the dataset.
- 3) Measures of Spread: I will calculate and interpret the Range, Variance, Standard Deviation, and Interquartile Range of the dataset.
- 4) Measures of Position: I will calculate and interpret the Five-Number summary of the dataset.

Description:

The temperature values in degrees Fahrenheit during first 10 days of the month of February 2025 in the city of Ashland Va.

Presentation:

- 1) The histogram shows that the data is skewed right.
- 2) The histogram shows that during the first 10 days in the month of February 2025 in Ashland Virginia, temperatures ranging from 45 to 50 degrees Fahrenheit were the most frequently recorded.

Analysis:

Let x be the temperature values for the dates of February 1st through February 10th.

Measures of Center:**Mean:**

$$\Sigma x = 56 + 44 + 67 + 71 + 49 + 48 + 62 + 50 + 53 + 47 = 547^{\circ}\text{F}$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{547}{10} = 54.7^{\circ}\text{F}$$

The mean average temperature in the city of Ashland VA for the first ten days in the month of February 2025 is 54.7°F.

Median:

I am going to sort the data in ascending order.

44,47,48,49,50,53,56,62,67,71

$$\text{Median} = Q_2 = \frac{50+53}{2} = \frac{103}{2} = 51.5^{\circ}\text{F}$$

When the temperature values in the first 10 days of February 2025 are arranged in ascending order, the median value is 51.5°F in the city of Ashland, VA.

Mode:

In the dataset, there is no temperature value that occurred more frequently than any other value.

Midrange:

The minimum temperature is 44°F, the maximum temperature is 71°F.

$$\text{Midrange} = \frac{\text{minimum} + \text{maximum}}{2} = \frac{44 + 71}{2} = 57.5$$

During the first 10 days in the month of February 2025 the average temperature of the minimum and maximum temperatures in the city of Ashland, VA is 57.5°F.

Measures of Spread:

I am going to calculate the range.

Range:

$$\text{Range} = \text{maximum} - \text{minimum}$$

$$= 71 - 44$$

$$= 27$$

The difference between the maximum temperature and the minimum temperature in the city of Ashland, VA during the first 10 days in the month of February 2025 is 27°F.

Variance:

x	$x - \bar{x}$	$(x - \bar{x})^2$
56	1.3	1.69
44	-10.7	144.49
67	12.3	151.29
71	16.3	265.69
49	-5.7	32.49
48	-6.7	44.89
62	7.3	53.29
50	-4.7	22.09
53	-1.7	2.89
47	-7.7	59.29
$\Sigma(x - \bar{x})^2 = 748.1$		

$$\text{Variance}, s^2 = \frac{\Sigma(x-\bar{x})^2}{n-1} = \frac{748.1}{10-1} = 83.12222222$$

Standard Deviation:

Next, I will calculate the Standard deviation.

To calculate the Standard Deviation, you take the square root of the Variance.

$$\text{Standard Deviation}, s = \sqrt{\text{Variance}} = \sqrt{83.12222222} = 9.117138927$$

The value of the standard deviation shows there is considerable spread among the temperature values.

Interquartile Range:

Next, I will calculate the Interquartile Range. This is the distance between the Upper and Lower Quartiles.

Dataset in ascending order: 44,47,48,49,50,53,56,62,67,71

$$n = 10$$

$$\frac{1}{4} \times 10 = 2.5 \quad \text{Round up to 3}$$

3rd value = 48

Lower Quartile, $Q_1 = 3rd\ value = 48^\circ F$

$$\frac{3}{4} \times 10 = 7.5 \quad \text{Round up to 8}$$

8th value = 62

Upper Quartile, $Q_3 = 8th\ value = 62$

Interquartile Range, $IQR = Q_3 - Q_1$

$$IQR = 62 - 48 = 14$$

Interquartile Range: shows that the variation in 50% of the daily temperature values in Ashland VA, during the first 10 days in the month of February 2025 is $14^\circ F$

Measures of Position:

The five number summary of the dataset and their interpretations are:

Minimum = $44^\circ F$

The minimum temperature value during the first 10 days in the month of February 2025 in Ashland Va is 44 degrees Fahrenheit.

Lower Quartile, $Q_1 = 48^\circ\text{F}$

48 degrees Fahrenheit is in the 25th percentile of the temperature values during the first 10 days in the month of February 2025 in Ashland VA.

Middle Quartile = Median, $Q_2 = 51.5^\circ\text{F}$

51.5 degrees Fahrenheit is in the 50th percentile of the temperature values during the first 10 days in the month of February 2025 in Ashland VA.

Upper Quartile, $Q_3 = 62^\circ\text{F}$

62 degrees Fahrenheit is in the 75th percentile of the temperature values during the first 10 days in the month of February 2025 in Ashland VA.

Maximum = 71°F

71 degrees Fahrenheit is the maximum value during the first 10 days in the month of February 2025 in Ashland VA.

Citations (MLA9)

“Climate.” *National Weather Service*, Department Of Commerce, :
www.weather.gov/wrh/Climate?wfo=akq. Accessed 28 Apr. 2025.

Chukwuemeka, Samuel. “Descriptive Statistics.” *Descriptive Statistics Project*,
<https://conferencepresentations.appspot.com/Projects/Statistics/DescriptiveStatistics.html>. Accessed 28 Apr. 2025.